

# Map Enumeration

## A Problem-Solving Exposition

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### Conventions

- All the graphs we consider can have loops and multiple edges.
- Unless otherwise stated, our graphs are connected.

**Definition 0.1** (Half-Edge [1]). We can think of an edge as a pair of half-edges, matched together. This helps to avoid some difficulties dealing with multiple edges and loops.

**Definition 0.2** (Corner [2]). A corner is the angular sector delimited by two consecutive edges around a vertex.

$\vdots$  {#def-degree of a vertex}

### Degree of a Vertex

The degree of a vertex  $v$  in a graph  $G$  is the number of half-edges incident to  $v$ .

$\vdots$

**Definition 0.3.**